TravelGo: A Cloud-Powered Real-Time Travel Booking Platform Using AWS

Project Description:

TravelGo is a full-stack, cloud-based travel booking platform designed to simplify the process of reserving buses, trains, flights, and hotels through a unified interface. Built using Flask as the backend framework, the application is deployed on Amazon EC2 and leverages DynamoDB for efficient storage of user data and bookings. TravelGo allows users to register, log in, search for transportation and accommodation options, and book their travel with ease. Once a booking is confirmed or cancelled, users receive real-time email notifications powered by AWS Simple Notification Service (SNS), keeping them informed throughout their journey.

Scenario 1: Hassle-Free Multi-Mode Travel Booking Experience

TravelGo offers users a unified platform to search and book buses, trains, flights, and hotels all in one place. For instance, a user planning a trip from Hyderabad to Bangalore can log in, select their preferred mode of transport, choose from available options, and proceed to booking. Flask manages the backend operations such as retrieving travel listings and processing user input in real-time. Hosted on AWS EC2, the platform remains responsive even during high-traffic hours like weekends or holiday seasons, allowing multiple users to browse and book without delay.

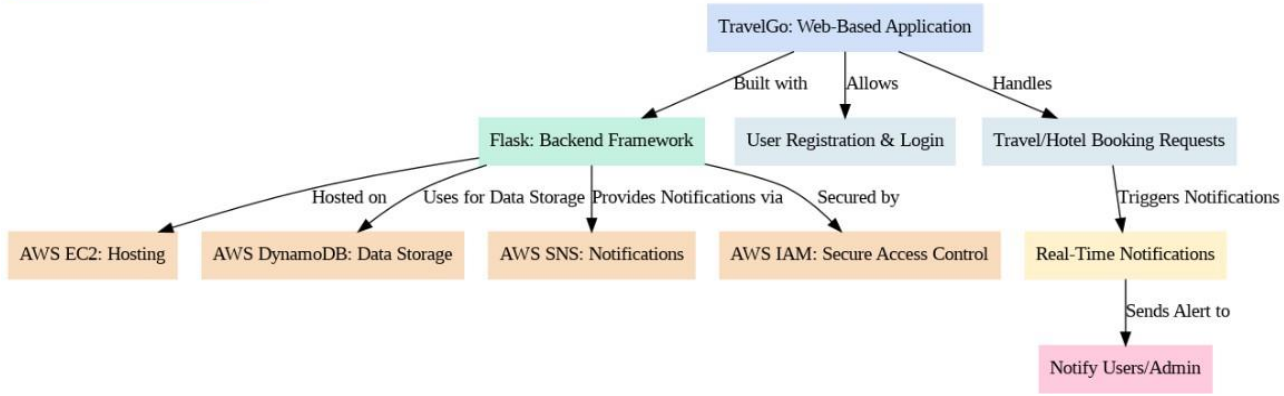
Scenario 2: Real-Time Booking Confirmation with AWS SNS

Once a booking is made—whether it's a train ticket or a hotel stay—TravelGo uses AWS SNS to instantly notify the user. For example, after a student books a hotel in Chennai, SNS sends a real-time email notification confirming the booking with all the relevant details. This notification is triggered from the Flask backend after the booking is successfully recorded in DynamoDB. Additionally, SNS can alert admin or service providers, ensuring transparency and real-time updates on every transaction.

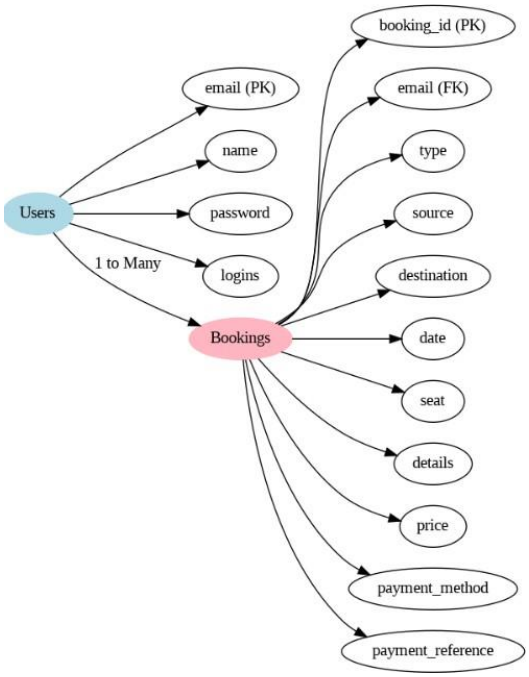
Scenario 3: Dynamic Dashboard with Personal Travel History

TravelGo features a dynamic user dashboard that displays all past and upcoming bookings for the logged-in user. For example, a user who has booked a flight and a hotel can view these bookings categorized by type, along with dates, price, and cancellation options. Flask fetches this data from AWS DynamoDB, which persistently stores all user bookings. The dashboard UI, powered by responsive HTML/CSS and Flask templates, ensures users can review or manage bookings anytime, from any device, with real-time updates and quick cancellation workflows supported.

AWS ARCHITECTURE



Entity Relationship (ER)Diagram:



Pre-requisites:

1. .**AWS Account Setup**: [AWS Account Setup]
2. **Understanding IAM**: [IAM Overview]
3. **Amazon EC2 Basics**: [EC2 Tutorial]
4. **DynamoDB Basics**: [DynamoDB Introduction]
5. **SNS Overview**: [SNS Documentation]
6. **Git Version Control**: [Git Documentation]

Project WorkFlow:

1. AWS Account Setup and Login

Activity 1.1: Set up an AWS account if not already done.

Activity 1.2: Log in to the AWS Management Console

2. DynamoDB Database Creation and Setup

Activity 2.1: Create a DynamoDB Table.

Activity 2.2: Configure Attributes for travel-Users and Bookings Tables.

3. SNS Notification Setup

Activity 3.1: Create an SNS Topic named BookingConfirmation for sending notifications.

Activity 3.2: Subscribe email to the topic to receive booking and cancellation alerts.

4. Backend Development and Application Setup

Activity 4.1:Develop the Backend Using Flask.

Activity 4.2: Integrate AWS Services Using boto3.

5. IAM Role Setup

Activity 5.1: Create IAM Role

Activity 5.2: Attach Policies

6. EC2 Instance Setup

Activity 6.1: Launch an EC2 instance to host the Flask application.

Activity 6.2: Configure security groups for HTTP, and SSH access.

7. Deployment on EC2

Activity 7.1:Upload Flask Files

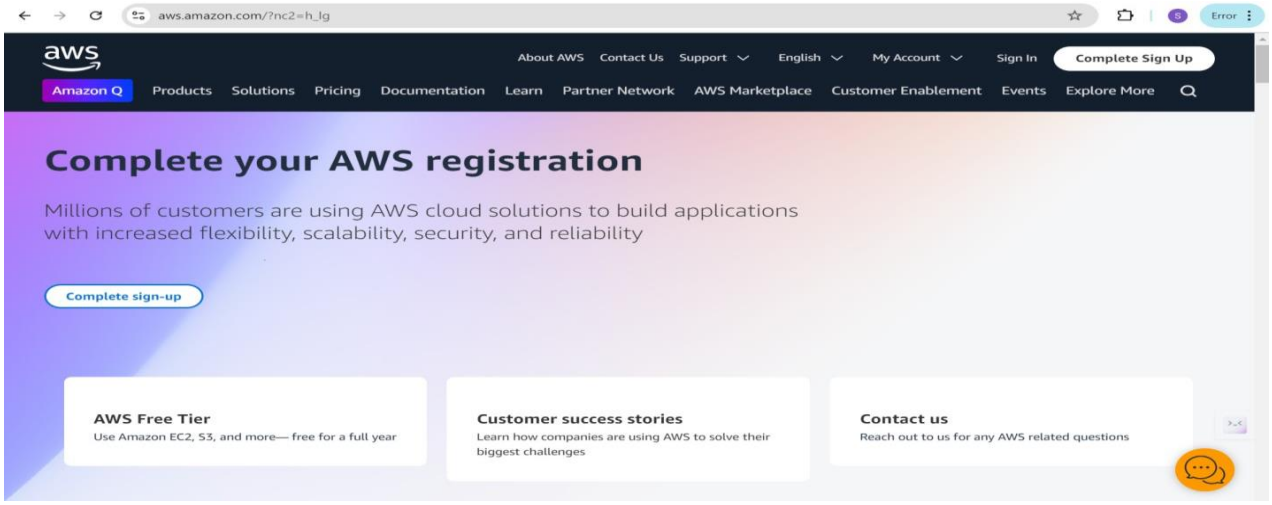
Activity 7.2: Run the Flask App

8. Testing and Deployment

Activity 8.1: Conduct functional testing to verify user registration, login, book requests, and notifications.

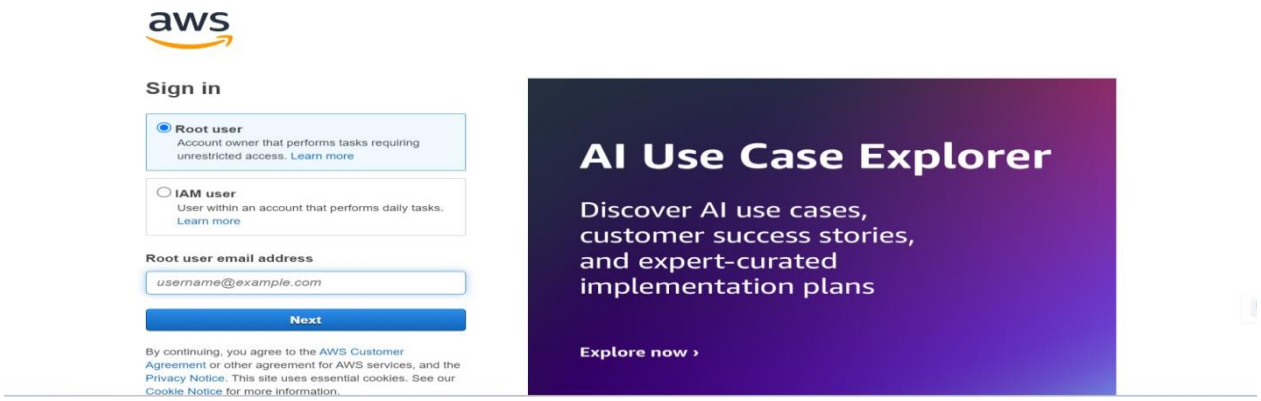
Milestone 1: AWS Account Setup and Login

- **Activity 1.1: Set up an AWS account if not already done.**
  - Sign up for an AWS account and configure billing settings.



- **Activity 1.2: Log in to the AWS Management Console**

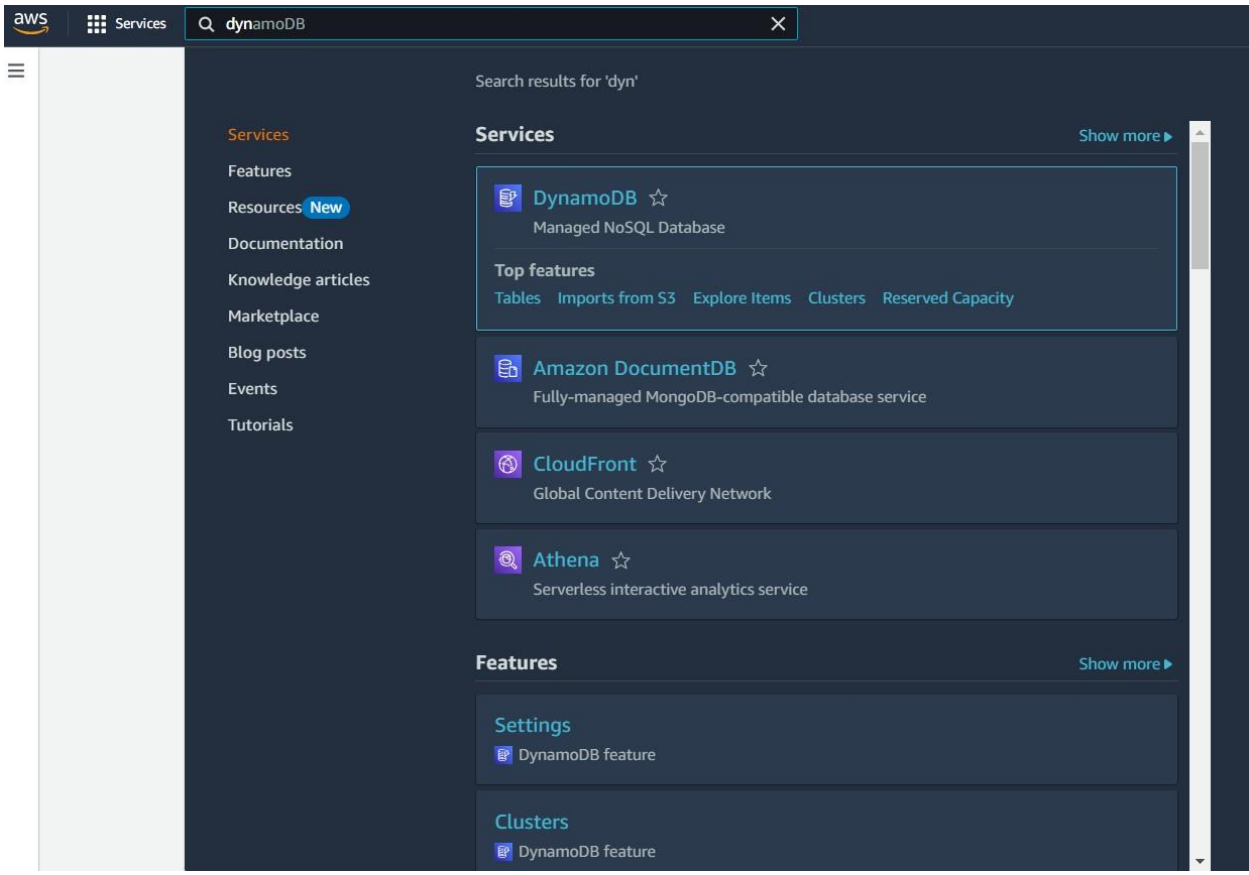
  - After setting up your account, log in to the AWS Management Console.

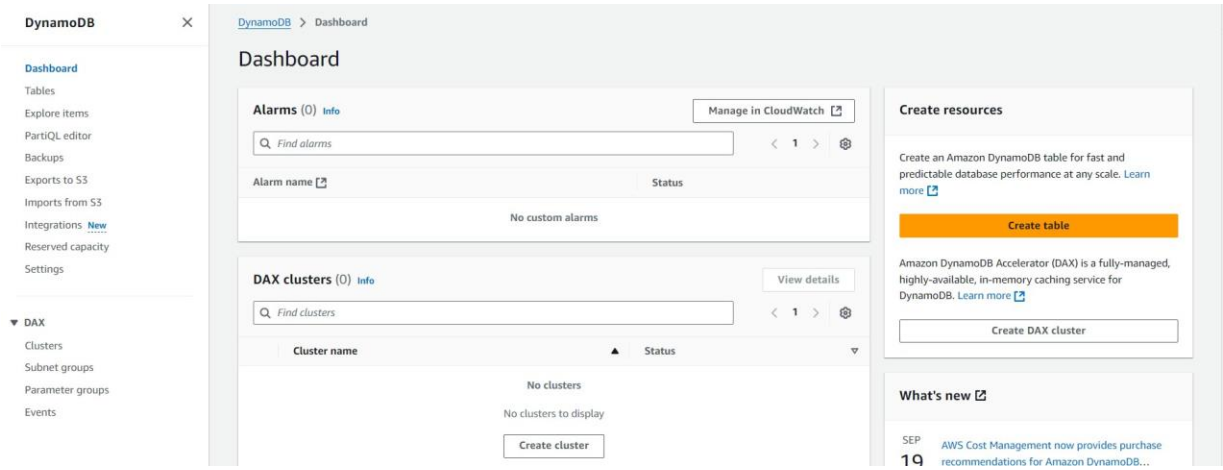


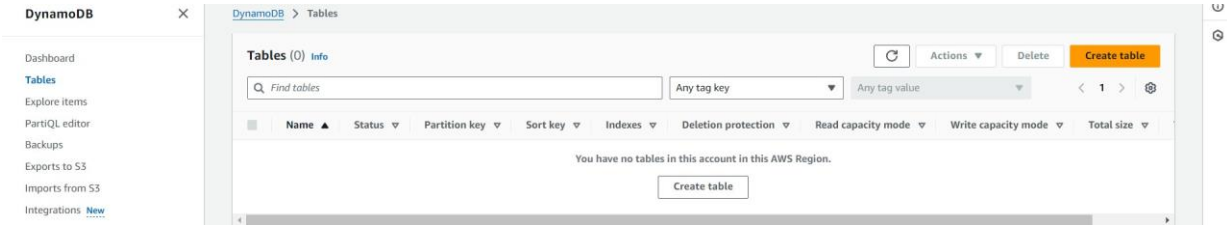
Milestone 2: DynamoDB Database Creation and Setup

- **Activity 2.1:Navigate to the DynamoDB**

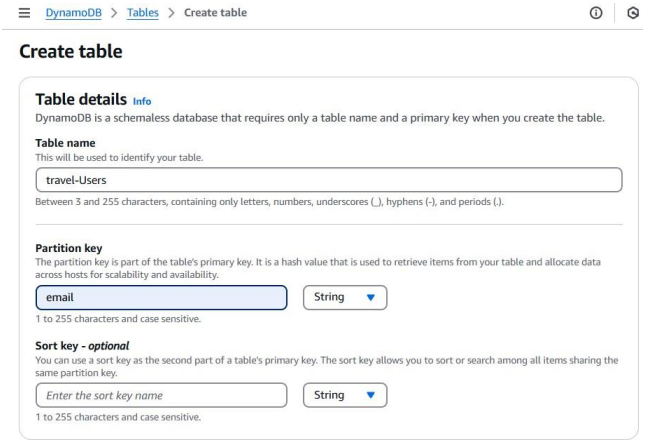
- In the AWS Console, navigate to DynamoDB and click on create tables.

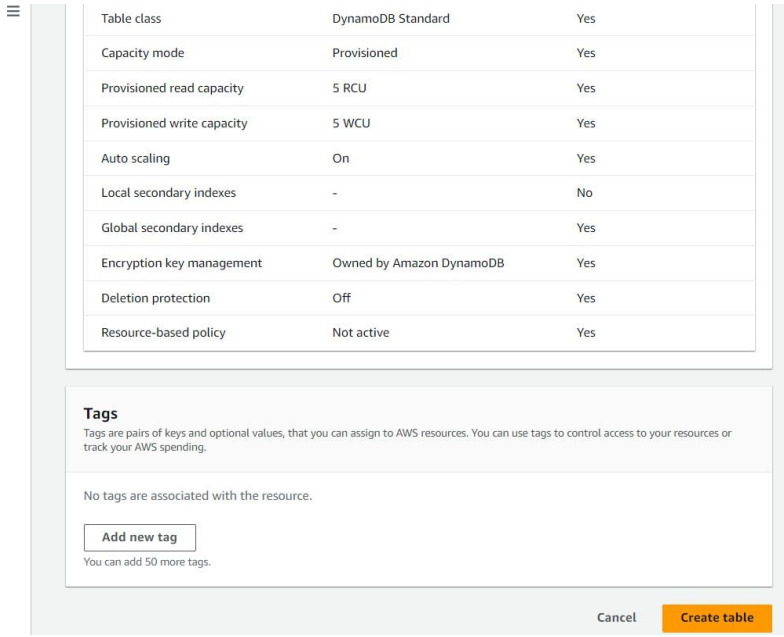




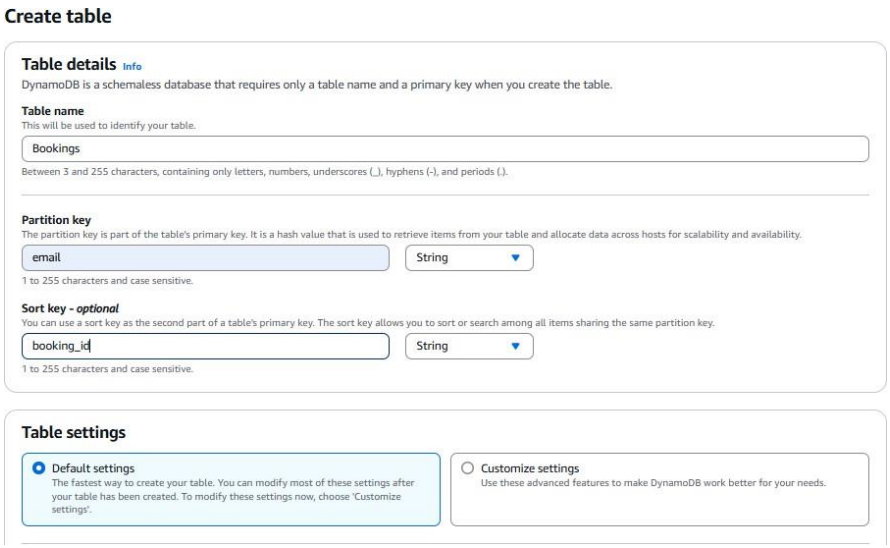


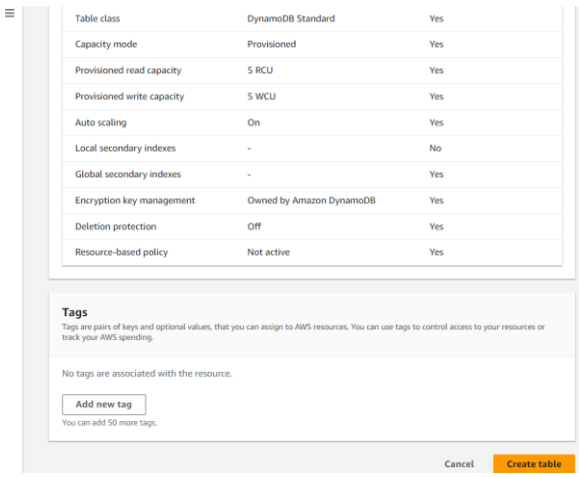
- **Activity 2.2:Create DynamoDB tables for storing registration details and booking records.**
    - Create a travel-Users table for storing registered user details with partition key "Email" with type String and click on create tables.

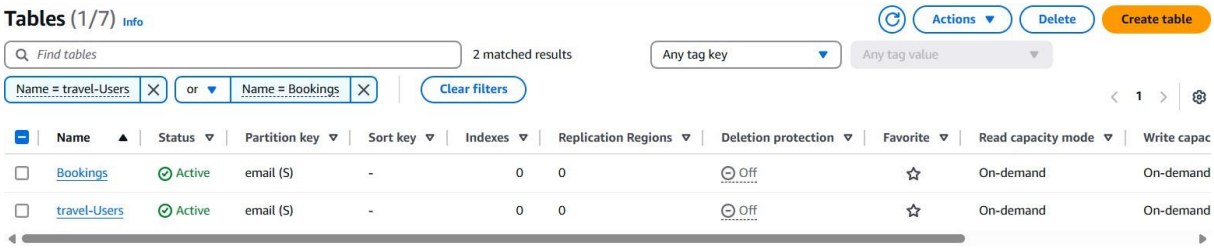




- Follow the same steps to create a `Bookings` for storing booking records with `email` as the partition key and `booking_id` as the sort key.

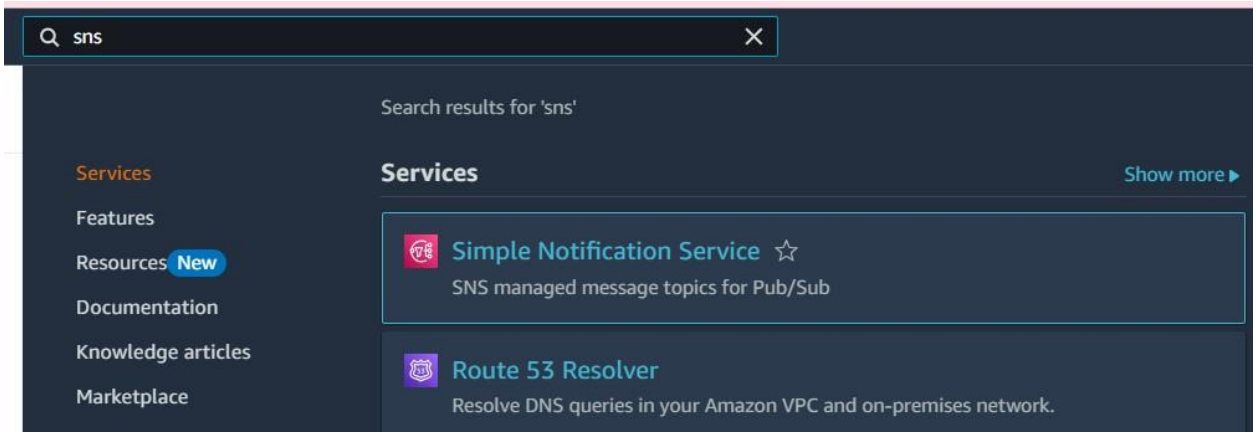


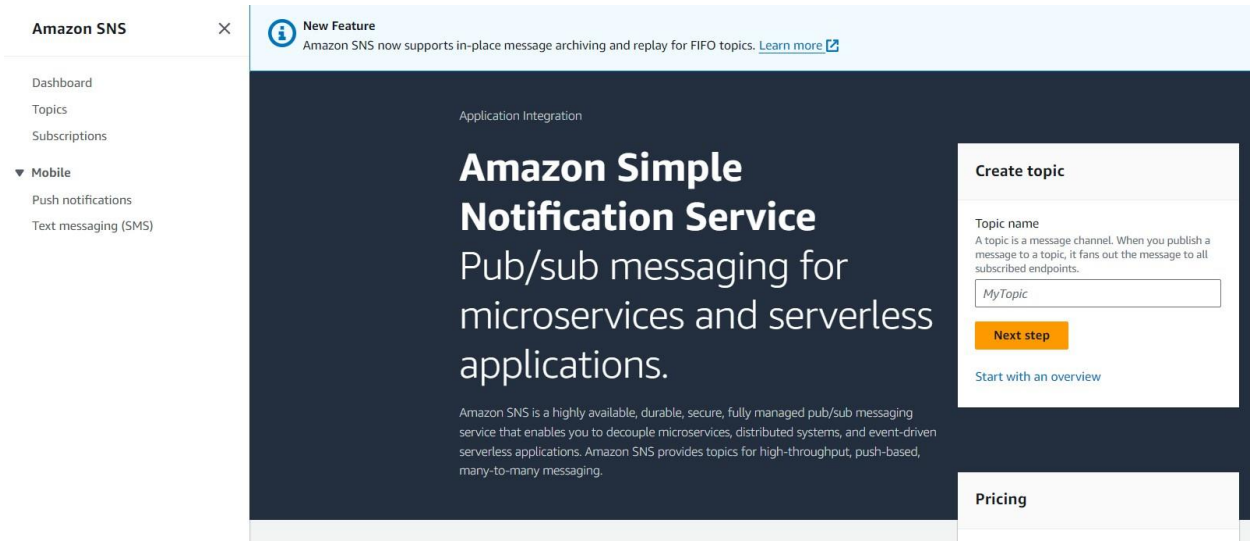




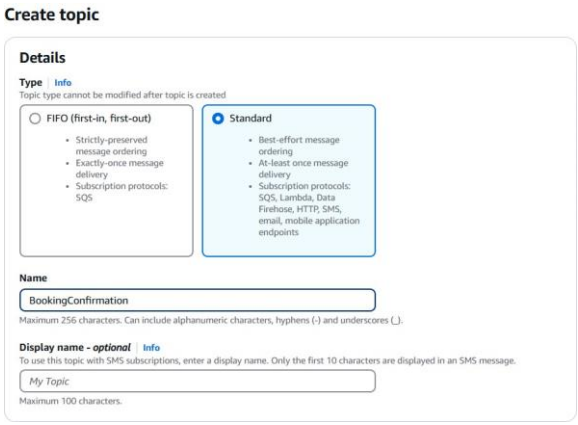
Milestone 3: SNS Notification Setup

- **Activity 3.1: Create an SNS Topic named BookingConfirmation for sending notifications.**
    - In the AWS Console, search for SNS and navigate to the SNS Dashboard.

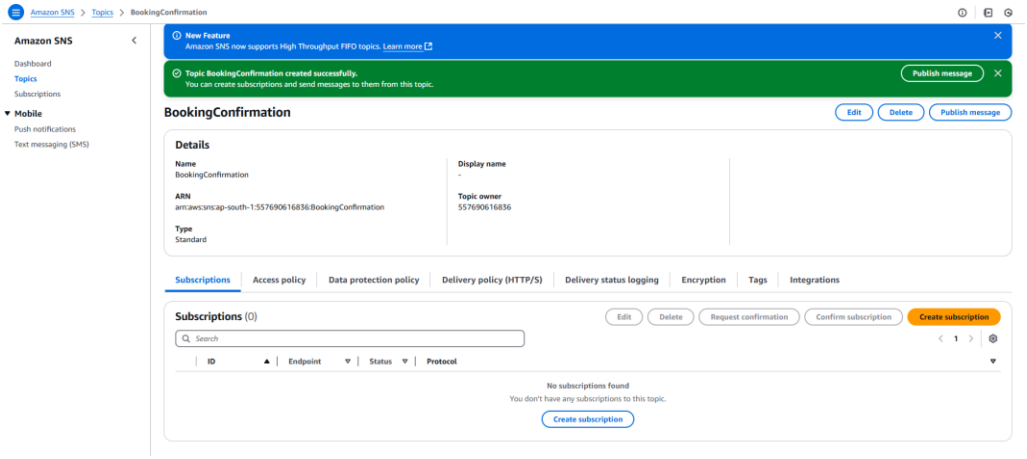




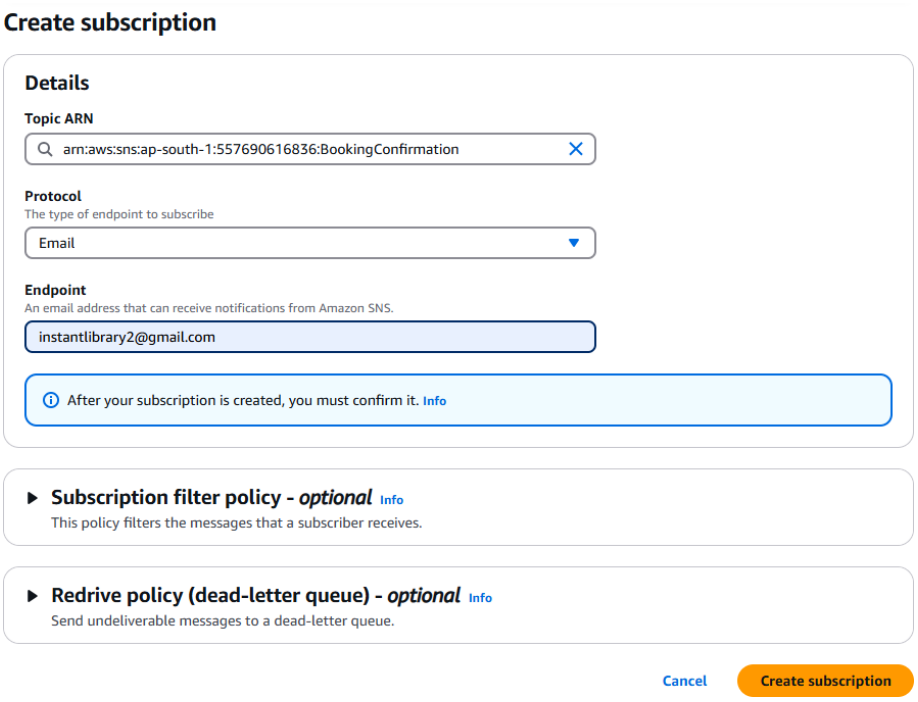
- Choose Standard type for general notification use cases and Click on Create Topic.

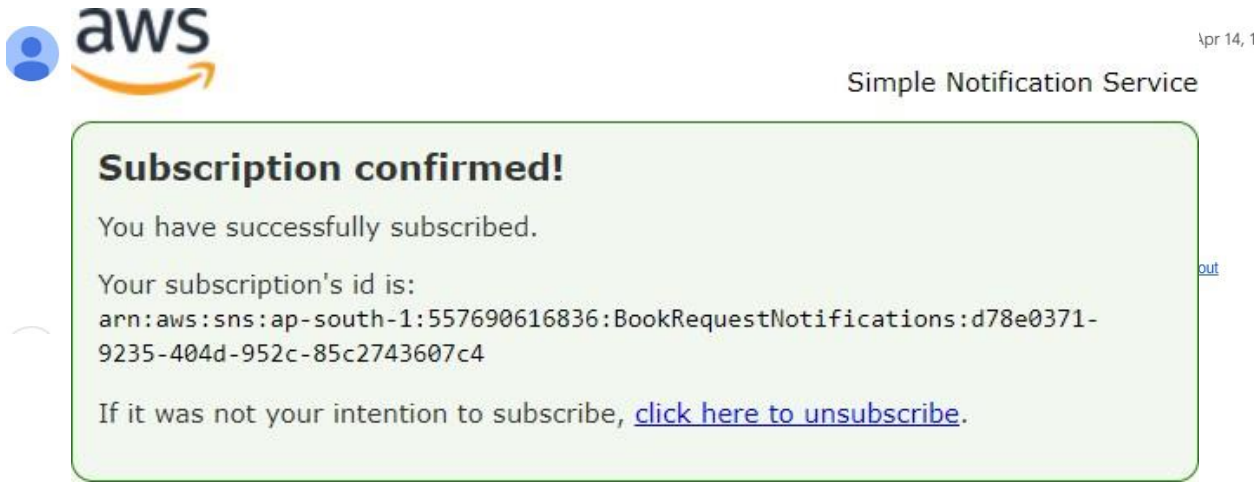


- Configure the SNS topic and note down the **Topic ARN**.



- **Activity 3.2: Subscribe email to the topic to receive booking and cancellation alerts.**
    - Subscribe users (or admin staff) to this topic via Email. When a book request is made, notifications will be sent to the subscribed emails.

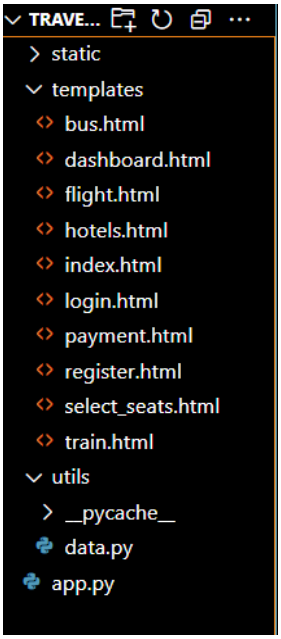




- Navigate to the subscribed Email account and Click on the confirm subscription in the AWS Notification- Subscription Confirmation mai

Milestone 4:Backend Development and Application Setup

- **Activity 4.1: Develop the backend using Flask**
  - File Explorer Structure:



Description: Create a `templates` folder containing HTML files for each major feature—bus, train, flight, hotels, seat selection, registration, login, payment, and dashboard—and manage backend logic in `app.py` and `utils/data.py`.

Description of the code :

- **Flask App Initialization**

  Import Libraries:

```
from flask import Flask, render_template, request, redirect, session
import uuid
import boto3
import os
from decimal import Decimal
from boto3.dynamodb.conditions import Attr
from utils.data import transport_data, hotel_data
```

Description: Import necessary Flask modules for web routing, Boto3 for AWS DynamoDB access, UUID for unique IDs, and custom transport and hotel data from `utils.data`

Initialize flask app:

```
app = Flask(__name__)
```

Description: Initialize the Flask application by creating an instance of the `Flask` class with the current module name.

DynamoDB Connection:

```
# AWS Setup
dynamodb = boto3.resource('dynamodb', region_name='ap-south-1')


# Tables
users_table = dynamodb.Table('travel-Users')
bookings_table = dynamodb.Table('Bookings')
```

Description: Use **boto3** to connect to **DynamoDB** for handling travel-users, booking database operations and also mention region_name where Dynamodb tables are created.

SNS Connection:

```
#sns setup

sns = boto3.client('sns', region_name='ap-south-1')
```

Description: Configure **SNS** to send notifications when a booking request is submitted.

- **Routes for Web Pages**

Register User:

```
@app.route('/')
def home():
    return render_template('index.html', logged_in='user' in session)
```

```
@app.route('/register', methods=['GET', 'POST'])
def register():
    if request.method == 'POST':
        email = request.form['email']
        response = users_table.get_item(Key={'email': email})
        if 'Item' in response:
            return render_template('register.html', message="User already
        users_table.put_item(Item={
            'email': email,
            'name': request.form['name'],
            'password': request.form['password'],
            'logins': 0
        })
        return redirect('/login')
    return render_template('register.html')
```

Description: Collecting registration data, and stores user details in the database.

- **login Route (GET/POST)**:

```
@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        email = request.form['email']
        password = request.form['password']
        response = users_table.get_item(Key={'email': email})
        user = response.get('Item')
        if user and user['password'] == password:
            session['user'] = email
            users_table.update_item(
                Key={'email': email},
                UpdateExpression="ADD logins :inc",
                ExpressionAttributeValues={':inc': Decimal(1)}
            )
            return '''
                <script>
                    localStorage.setItem("loggedIn", "true");
                    window.location.href = "/dashboard";
                </script>
            '''
        return render_template('login.html', message="Invalid credentials.
    return render_template('login.html')
```

Description: Verifies user credentials, increments login count, and redirects to the dashboard on success.

Dashboard route:

```
@app.route('/dashboard')
def dashboard():
    if 'user' not in session:
        return redirect('/login')
    email = session['user']
    user = users_table.get_item(Key={'email': email}).get('Item')
    response = bookings_table.scan(FilterExpression=Attr('user_email').eq(email))
    return render_template('dashboard.html', name=user['name'], bookings=response['Items'])
```

Description: Define the `/dashboard` route to display the user's booking data from DynamoDB after verifying the session and fetching user details using their email.

Transport Routes:

```
# ---------- Transport Pages ----------
@app.route('/bus', methods=['GET', 'POST'])
def bus():
    if request.method == 'POST':
        s, d, dt = request.form['source'], request.form['destination'], request.form['date']
        buses = [b for b in transport_data['bus'] if b['source'] == s and b['destination'] == d]
        return render_template('bus.html', buses=buses, source=s, destination=d, date=dt)
    return render_template('bus.html', buses=None)

@app.route('/train', methods=['GET', 'POST'])
def train():
    if request.method == 'POST':
        s, d, dt = request.form['source'], request.form['destination'], request.form['date']
        trains = [t for t in transport_data['train'] if t['source'] == s and t['destination'] == d]
        return render_template('train.html', trains=trains, source=s, destination=d, date=dt)
    return render_template('train.html', trains=None)

@app.route('/flight', methods=['GET', 'POST'])
def flight():
    if request.method == 'POST':
        s, d, dt = request.form['source'], request.form['destination'], request.form['date']
        flights = [f for f in transport_data['flight'] if f['source'] == s and f['destination'] == d]
        return render_template('flight.html', flights=flights, source=s, destination=d, date=dt)
    return render_template('flight.html', flights=None)

@app.route('/hotels', methods=['GET', 'POST'])
def hotels():
    if request.method == 'POST':
        city = request.form['city']
        hotels = [h for h in hotel_data if h['location'] == city]
        return render_template('hotels.html', hotels=hotels, city=city)
    return render_template('hotels.html', hotels=None)
```

```python
@app.route('/select_seats')
def select_seats():
    return render_template("select_seats.html")
```

Description: Define separate route handlers for bus, train, flight, and hotel search functionalities using POST requests to filter results from local datasets based on user input like source, destination, date, or city and also add seat selecting routes for bus page..

Booking routes:

```python
# ---------- Booking ----------
@app.route('/book', methods=['POST'])
def book():
    if 'user' not in session:
        return redirect('/login')

    session['pending_booking'] = {
        "booking_id": str(uuid.uuid4())[:8],
        "type": request.form['type'],
        "source": request.form['source'],
        "destination": request.form['destination'],
        "date": request.form['date'],
        "seat": request.form.get('seat', 'N/A'),
        "details": request.form['details'],
        "price": Decimal(request.form['price']),
        "user_email": session['user'],
        "email": session['user']
    }
    return render_template("payment.html", booking=session['pending_booking'])
```

Description: Saves booking details in session and redirects to the payment page if the user is logged in.

Payment Route:

```
@app.route('/payment', methods=['POST'])
def payment():
    if 'user' not in session or 'pending_booking' not in session:
        return redirect('/login')

    booking = session.pop('pending_booking')
    booking['payment_method'] = request.form['method']
    booking['payment_reference'] = request.form['reference']

    bookings_table.put_item(Item=booking)

    # SNS Booking Notification
    try:
        sns.publish(
            TopicArn=os.getenv('SNS_TOPIC_ARN'),
            Message=f"✅ Booking Confirmed!\n{booking['details']} on {booking['date']} for ₹{booking['price']}",
            Subject="TravelGo Booking Confirmation"
        )
    except Exception as e:
        print("SNS Error (booking):", e)

    return redirect('/dashboard')
```

Description: Processes payment details, stores the booking in the database, and sends an SNS confirmation notification before redirecting to the dashboard.

Remove Booking route:

```
@app.route('/remove_booking', methods=['POST'])
def remove_booking():
    if 'user' not in session:
        return redirect('/login')
    email = session['user']
    booking_id = request.form.get('booking_id')

    try:
        bookings_table.delete_item(Key={'email': email, 'booking_id': booking_id})
        # SNS Cancellation Notification
        sns.publish(
            TopicArn=os.getenv('SNS_TOPIC_ARN'),
            Message=f"❌ Booking Canceled\nBooking ID: {booking_id}",
            Subject="TravelGo Cancellation"
        )
    except Exception as e:
        print("Delete Error:", e)

    return redirect('/dashboard')
```

Description: Deletes a booking from the database and sends a cancellation notification via SNS, then redirects to the dashboard.

Logout route:

```
@app.route('/logout')
def logout():
    session.clear()
    return '''
        <script>
            localStorage.setItem("loggedIn", "false");
            window.location.href = "/";
        </script>
    '''
```

Description: Implement the logout route to clear session data and reset `loggedIn` in localStorage before redirecting the user to the homepage.

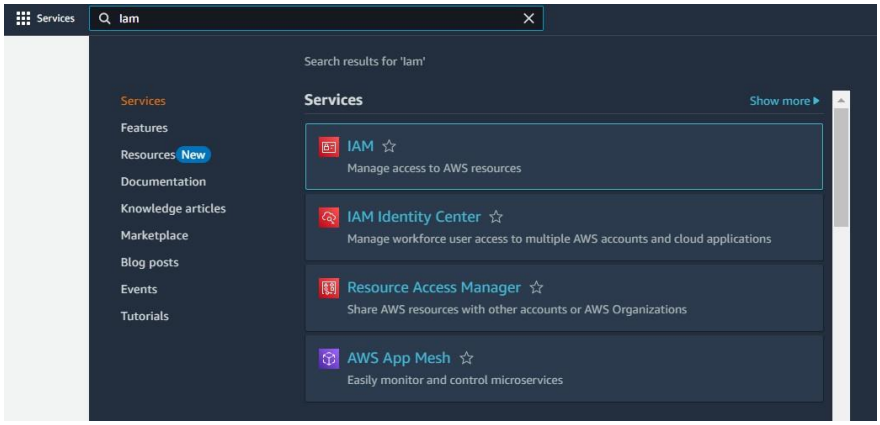
Deployment Code:

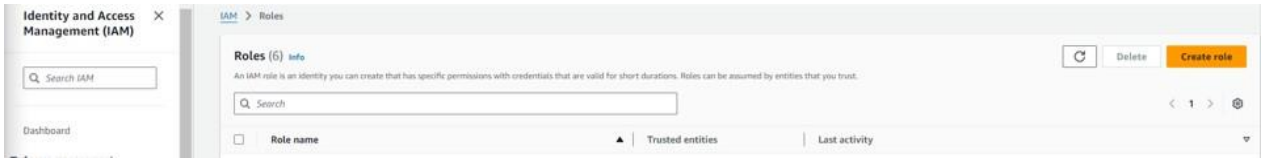
```
if __name__ == "__main__":
    app.run(host='0.0.0.0', port=80, debug=True)
```

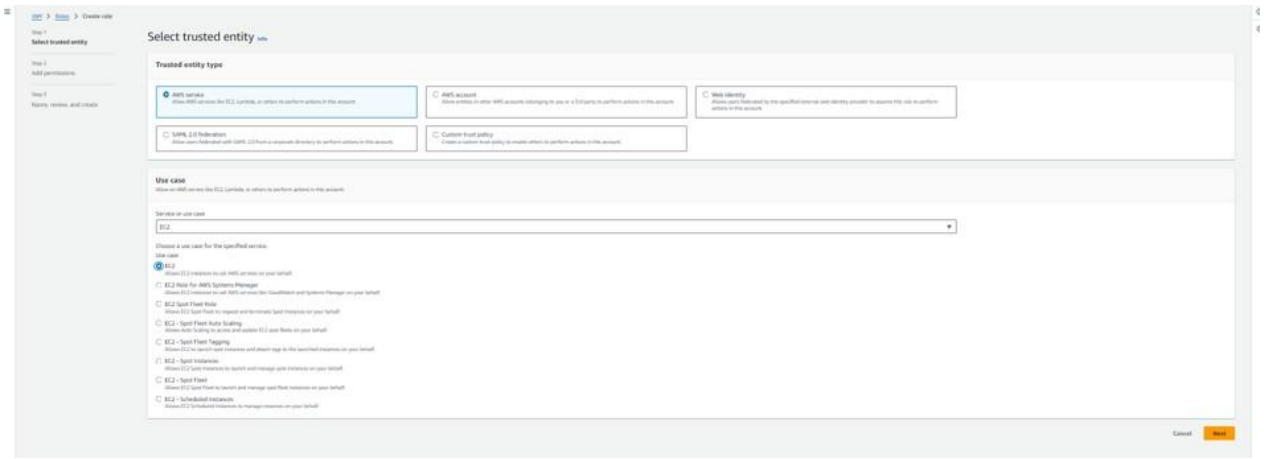
Description: Start the Flask application on all network interfaces at port 80 with debug mode enabled for easier development and testing.

Milestone 5: IAM Role Setup

- **Activity 5.1:Create IAM Role.**
    - In the AWS Console, go to IAM and create a new IAM Role for EC2 to interact with DynamoDB and SNS.



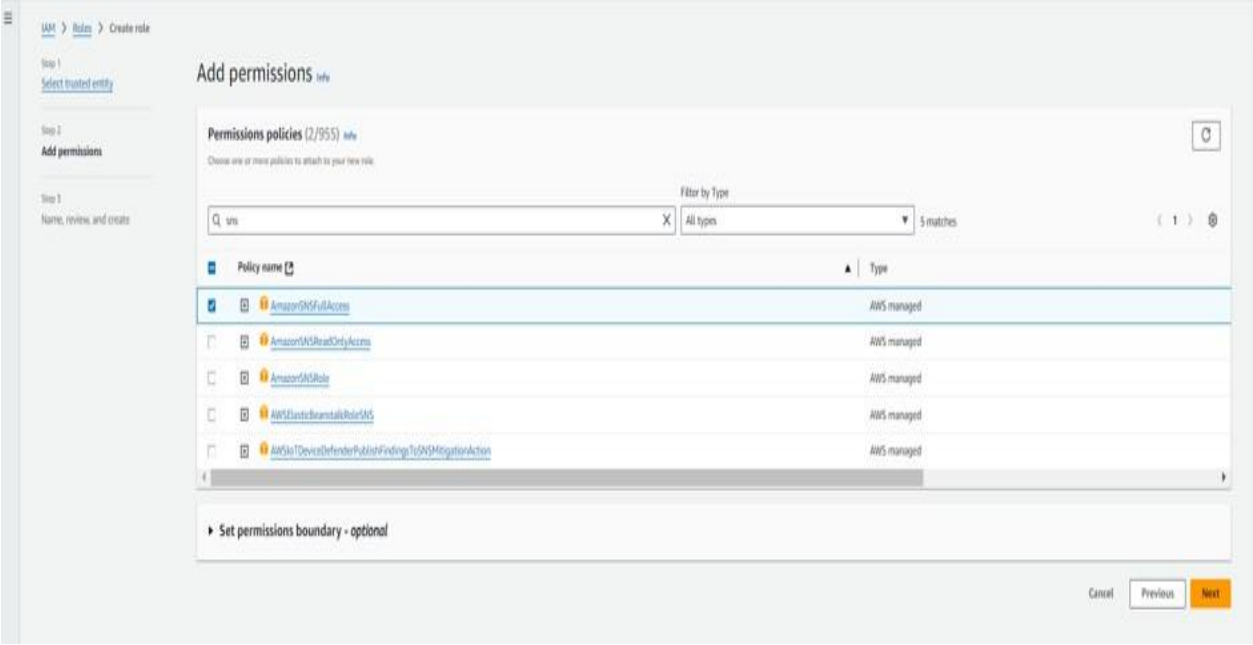


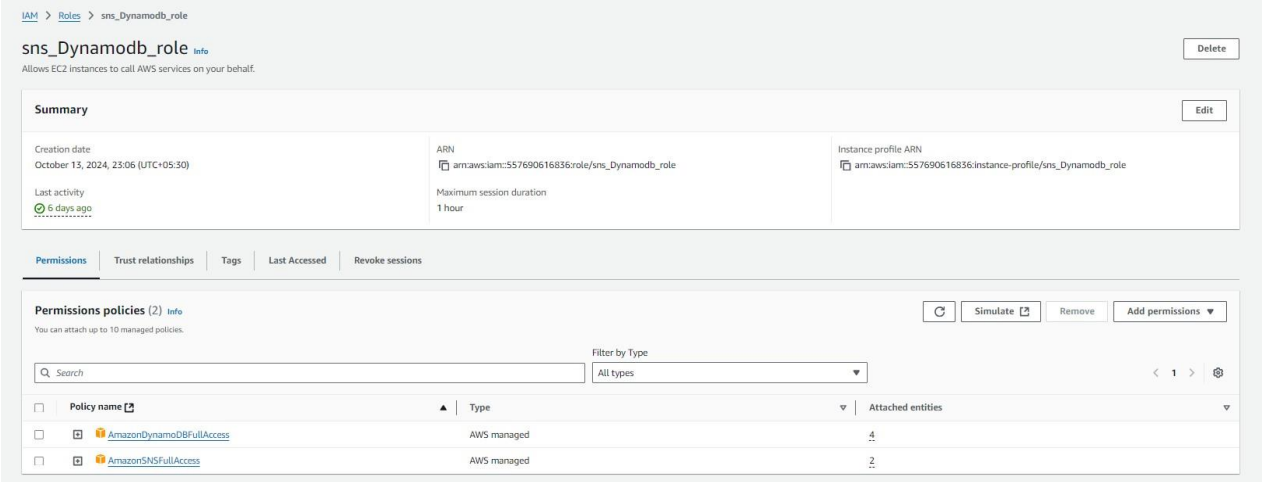


- **Activity 5.2: Attach Policies.**

  Attach the following policies to the role:

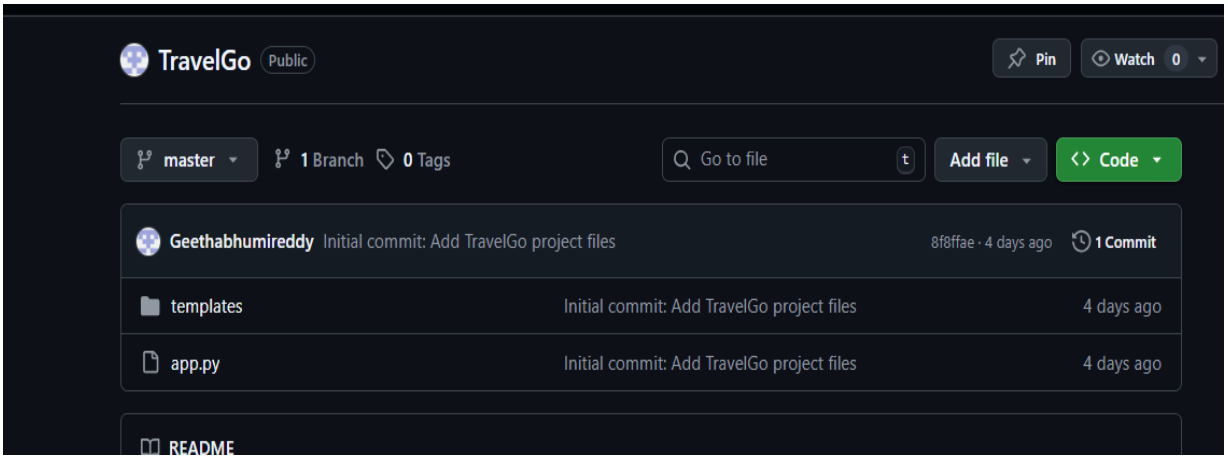
  - AmazonDynamoDBFullAccess: Allows EC2 to perform read/write operations on DynamoDB.
  - AmazonSNSFullAccess: Grants EC2 the ability to send notifications via SNS.

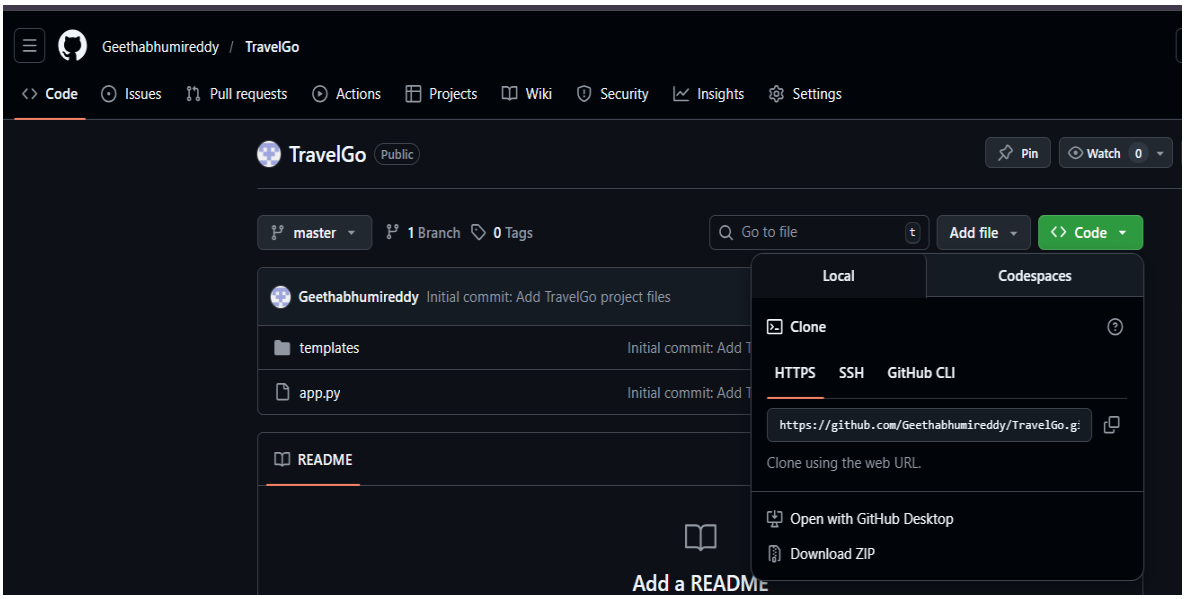

IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Add permissions Info
Permissions policies (2/955) Info
Choose one or more policies to attach to your new role.
Filter by Type
sns
All types
5 matches
Policy name
Type
AmazonSNSFullAccess
AWS managed
AmazonSNSReadOnlyAccess
AWS managed
AmazonSNSRole
AWS managed
AWSElasticBeanstalkRoleSNS
AWS managed
AWSIoTDeviceDefenderPublishFindingsToSNSMitigationAction
AWS managed
Set permissions boundary - optional
Cancel
Previous
Next



Milestone 6: EC2 Instance Setup

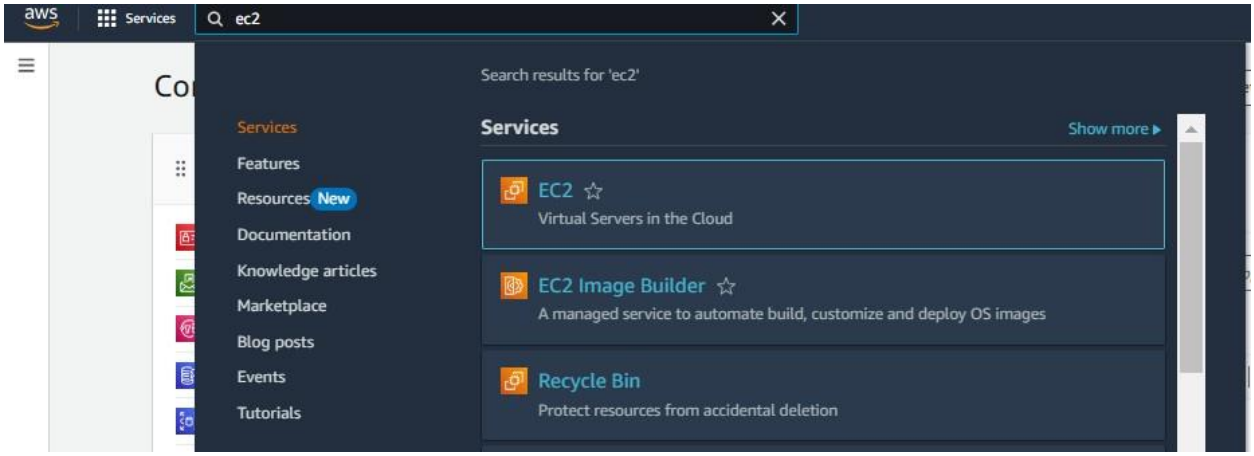
- **Note: Load your Flask app and Html files into GitHub repository.**



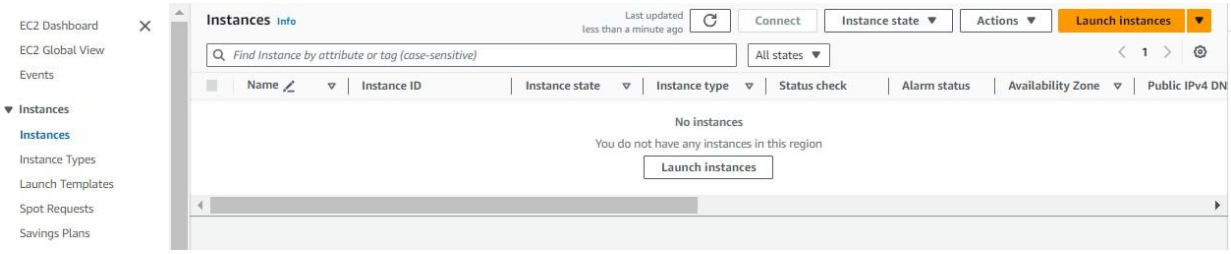


- **Activity 6.1: Launch an EC2 instance to host the Flask application.**

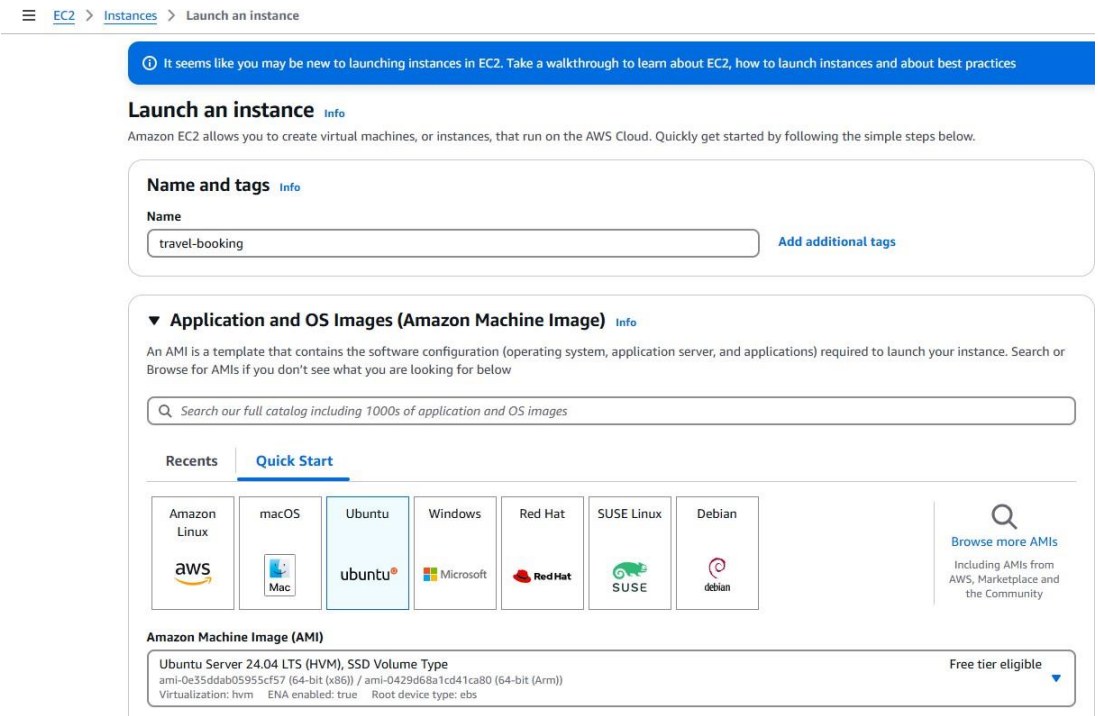
  - **Launch EC2 Instance**
    - In the AWS Console, navigate to EC2 and launch a new instance.



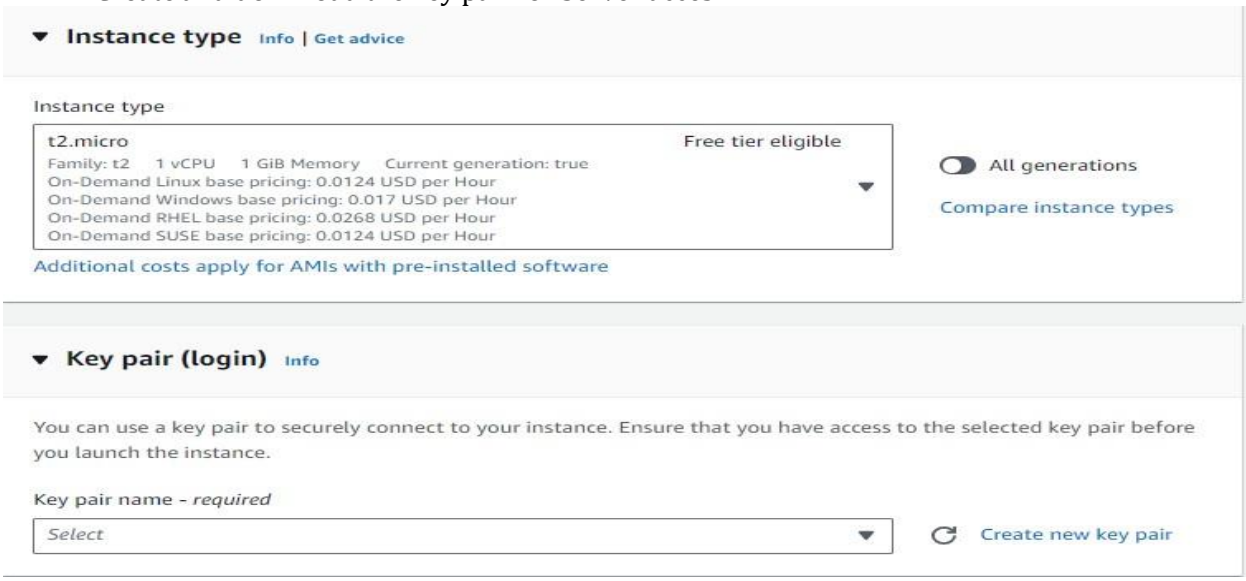
- Click on Launch instance to launch EC2 instance



- Choose Amazon Linux 2 or Ubuntu as the AMI and t2.micro as the instance type (free-tier eligible).

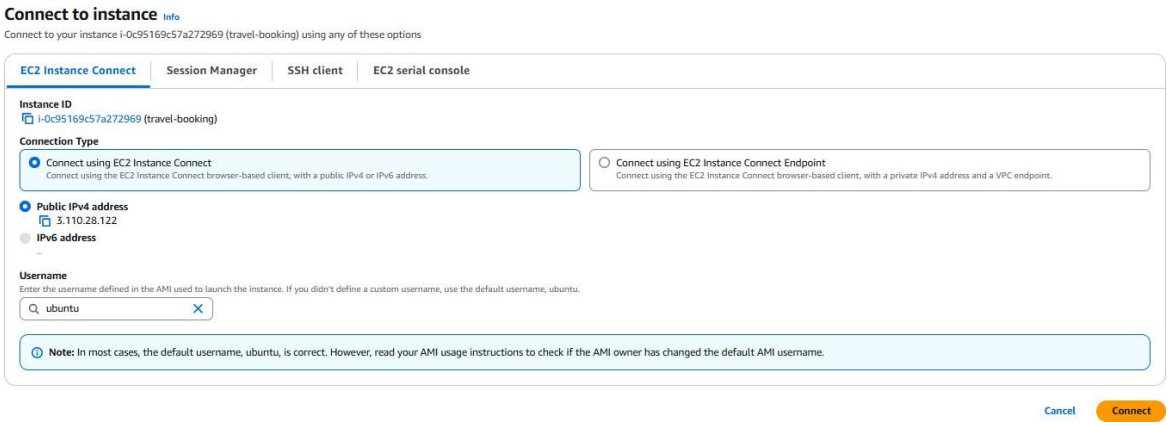


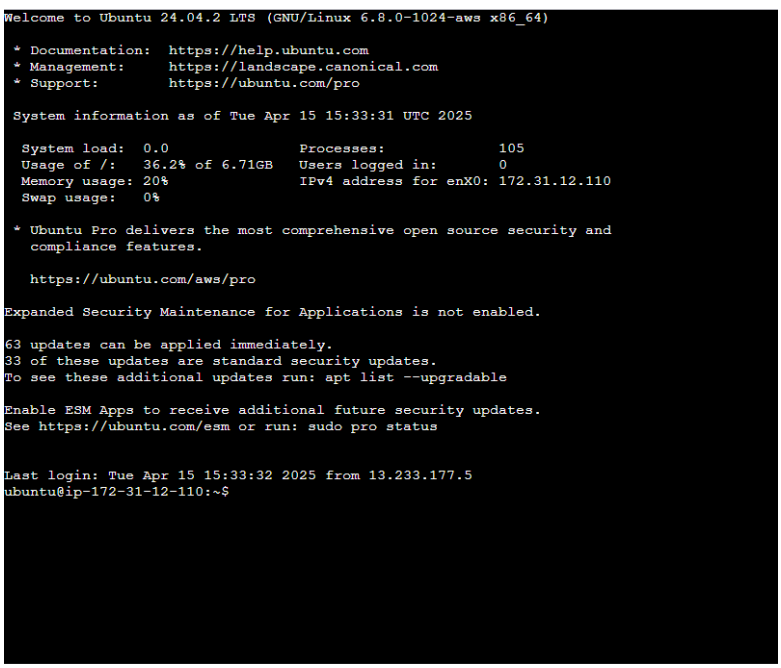
- Create and download the key pair for Server acces



- To connect to EC2 using **EC2 Instance Connect**, start by ensuring that an **IAM role** is attached to your EC2 instance. You can do this by selecting your instance, clicking on **Actions**, then navigating to **Security** and selecting **Modify IAM Role** to attach the appropriate role. After the IAM role is connected, navigate to the **EC2** section in the **AWS Management Console**. Select the **EC2 instance** you wish to connect to. At the top of the **EC2 Dashboard**, click the **Connect** button. From the connection methods presented, choose **EC2 Instance Connect**. Finally, click **Connect** again, and a new browser-based terminal will open, allowing you to access your EC2 instance directly from your browser.

- Now connect the EC2 with the files





i-0c95169c57a272969 (travel-booking)

PublicIPs: 13.203.222.149 PrivateIPs: 172.31.12.110

Milestone 7: Deployment on EC2

Activity 7.1: Install Software on the EC2 Instance

 # Update and install Python + pip

sudo apt update

sudo apt install python3-pip -y

sudo apt install git -y

sudo apt install python3-venv -y

python3 -m venv venv

source venv/bin/activate

Activity 7.2:Clone Your Flask Project from GitHub

Clone your project repository from GitHub into the EC2 instance using Git.

Run: 'git clone https://github.com/your-github-username/your-repository-name.git'

Note: change your-github-username and your-repository-name with your credentials

here: 'https://github.com/AlekhyaPenubakula/Travel-Booking-website_Ec2_Dynamodb_SNS.git'

- This will download your project to the EC2 instance.

To navigate to the project directory, run the following command:

cd <your-folder>

Once inside the project directory, configure and run the Flask application by executing the following command with elevated privileges:

Run the Flask Application

export SNS_TOPIC_ARN=arn:aws:sns:ap-south-1:557690616836:BookingConfirmation

git pull origin main (Only if you made changes in git and want to reflect them in the EC2 terminal.)

 # Install requirements

pip install flask boto3

Launch the Flask app:

sudo -E venv/bin/python3 app.py

Verify the Flask app is running:

http://your-ec2-public-ip

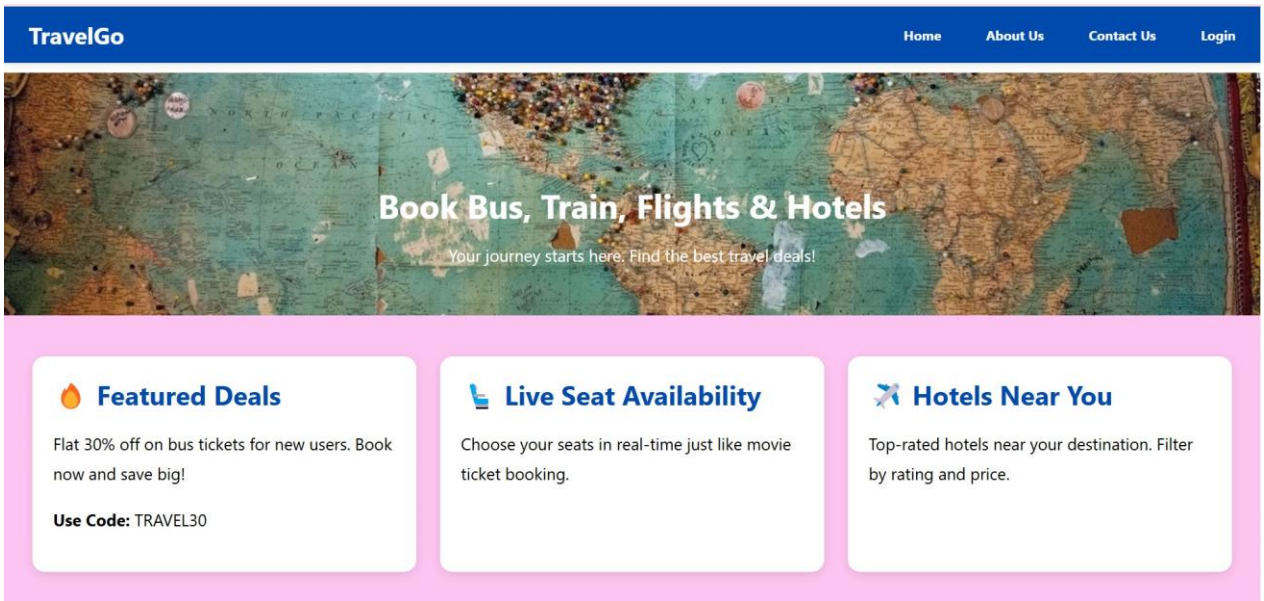
- Run the Flask app on the EC2 instance

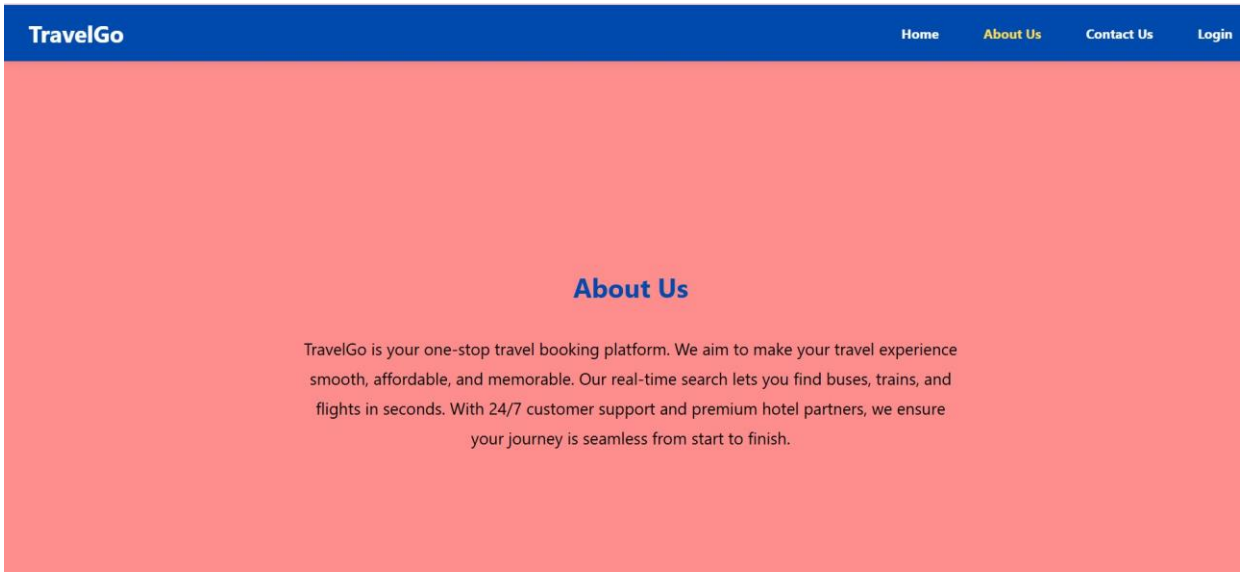
```
Last login: Mon Apr 14 16:27:44 2025 from 13.233.177.5
ubuntu@ip-172-31-12-110:~$ cd Travel-Booking-website_Ec2_Dynamodb_SNS
ubuntu@ip-172-31-12-110:~/Travel-Booking-website_Ec2_Dynamodb_SNS$ cd travel_booking_system
ubuntu@ip-172-31-12-110:~/Travel-Booking-website_Ec2_Dynamodb_SNS/travel_booking_system$ export SNS_TOPIC_ARN=arn:aws:sns:ap-south-1:557690616836:BookingConfirmation
ubuntu@ip-172-31-12-110:~/Travel-Booking-website_Ec2_Dynamodb_SNS/travel_booking_system$ git pull origin main
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.68 KiB | 429.00 KiB/s, done.
From https://github.com/AlekhyaPenubakula/Travel-Booking-website_Ec2_Dynamodb_SNS
 * branch            main       -> FETCH_HEAD
   2289f47..397e3a1  main       -> origin/main
Updating 2289f47..397e3a1
Fast-forward
 travel_booking_system/templates/hotels.html | 107 ++++++
```

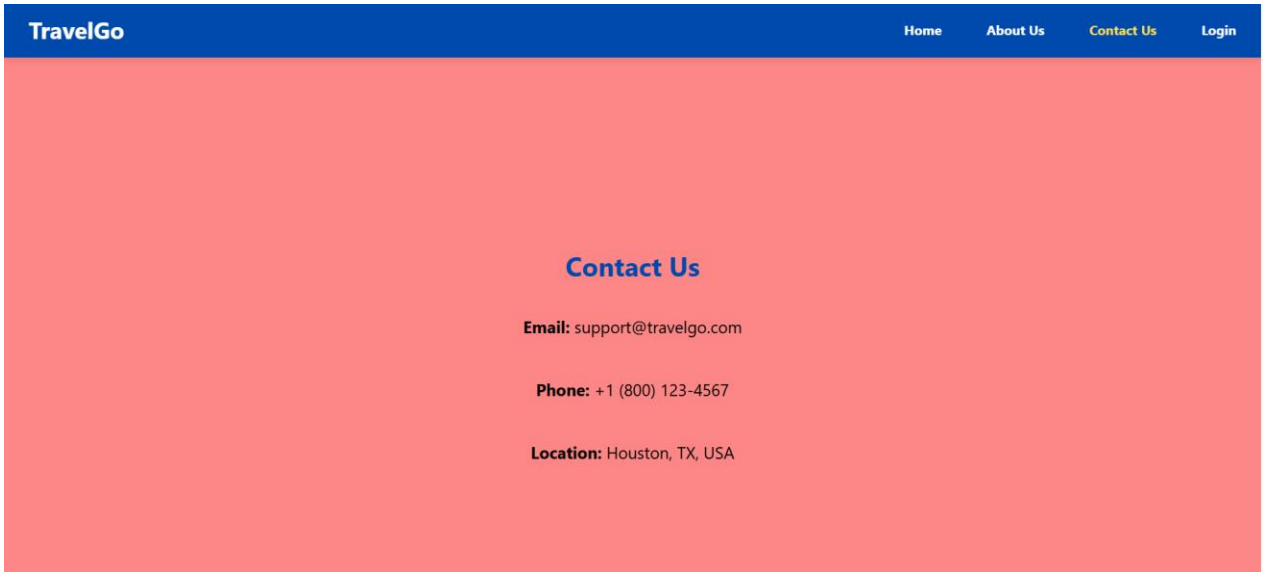
Milestone 8: Testing and Deployment

- **Activity 8.1: Conduct functional testing to verify user registration, login, book requests, and notifications.**

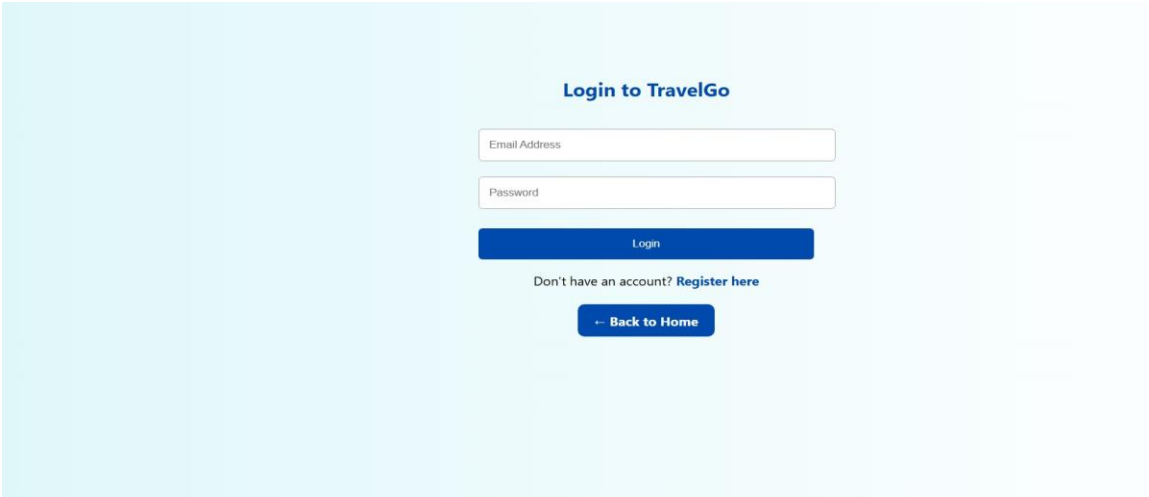
Home Page:



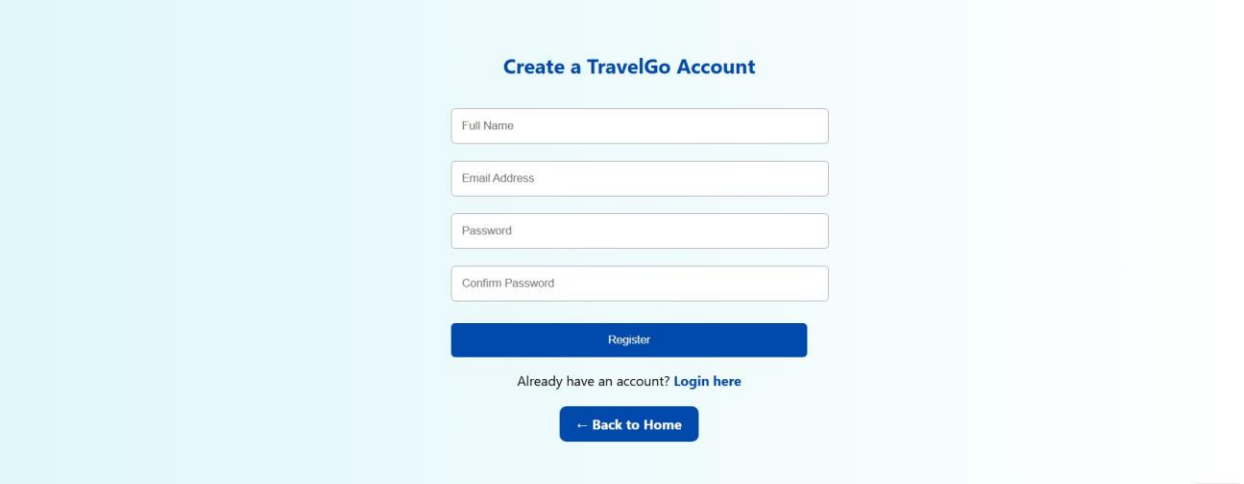


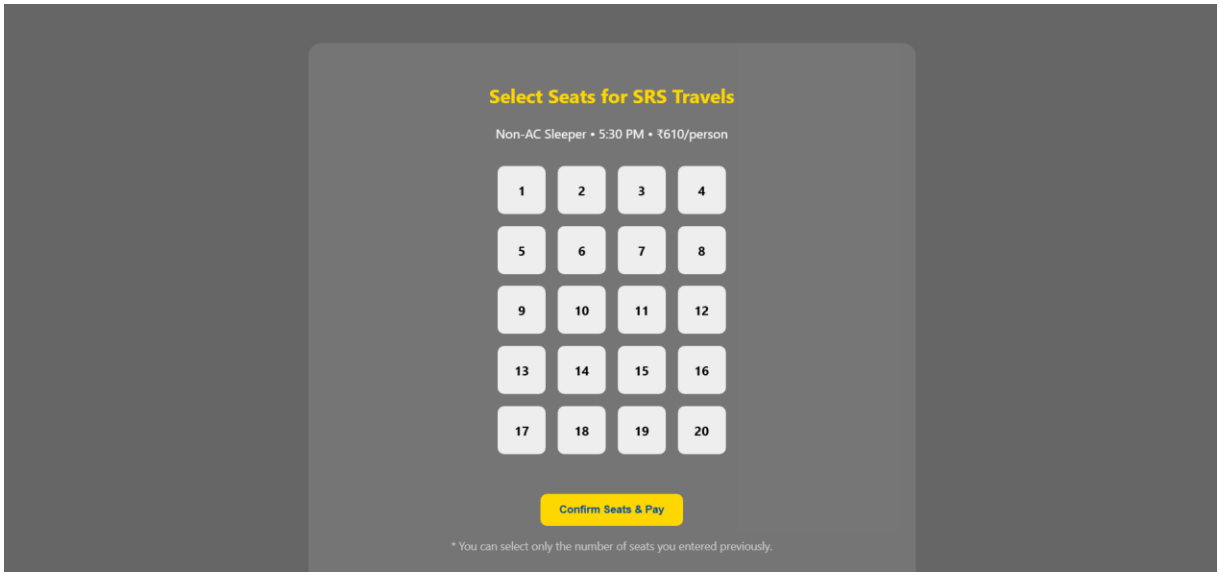


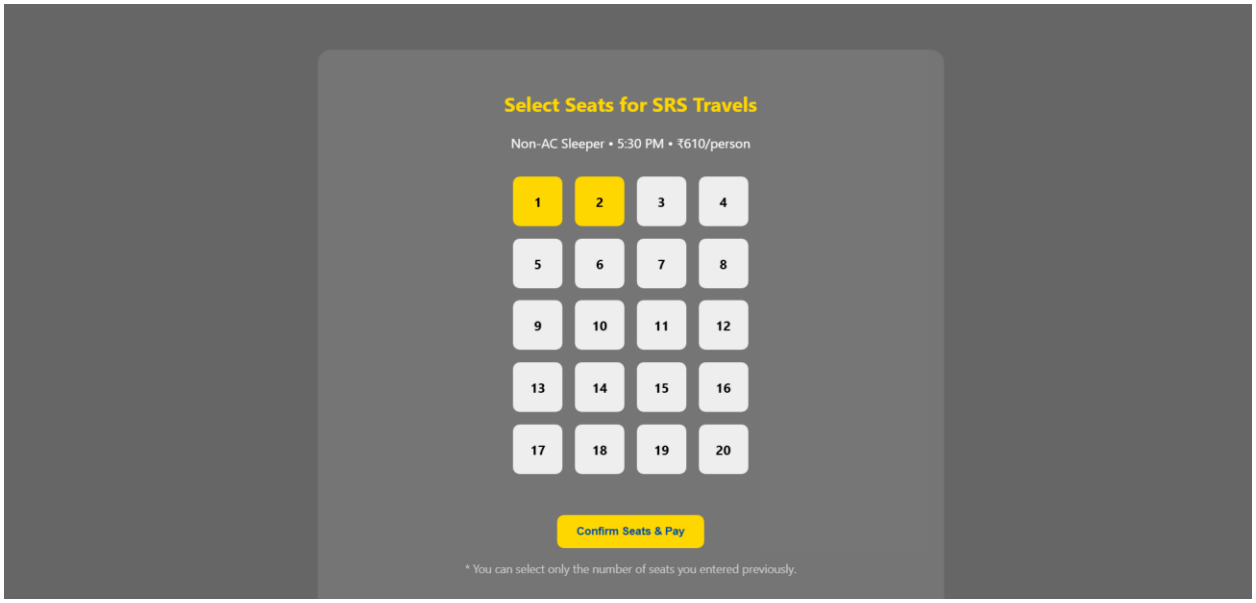
Login Page:



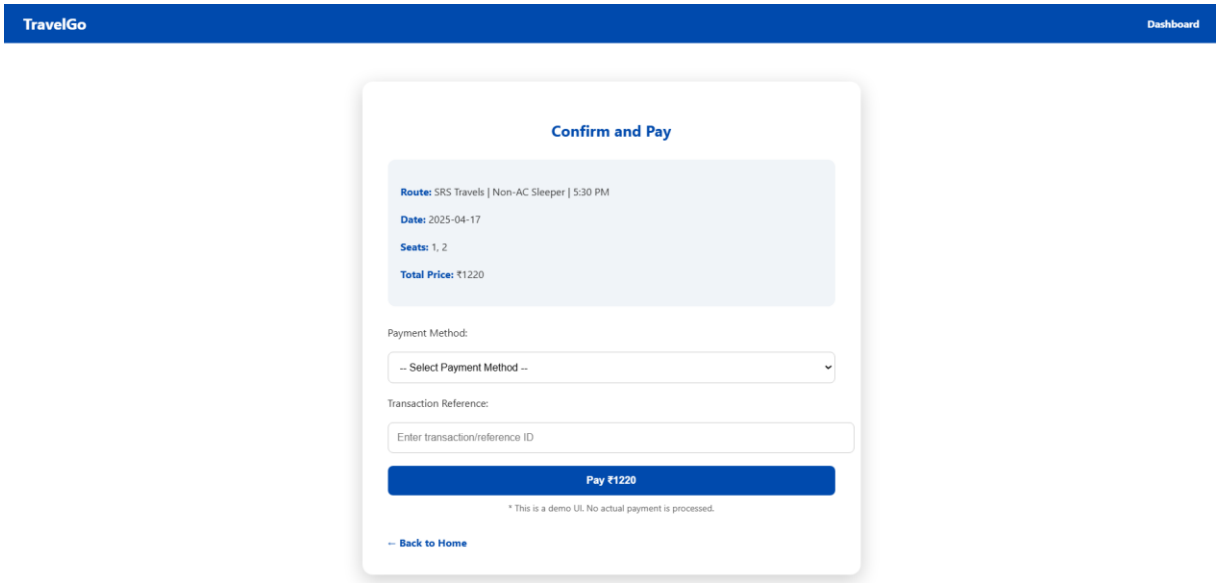
Register Page:

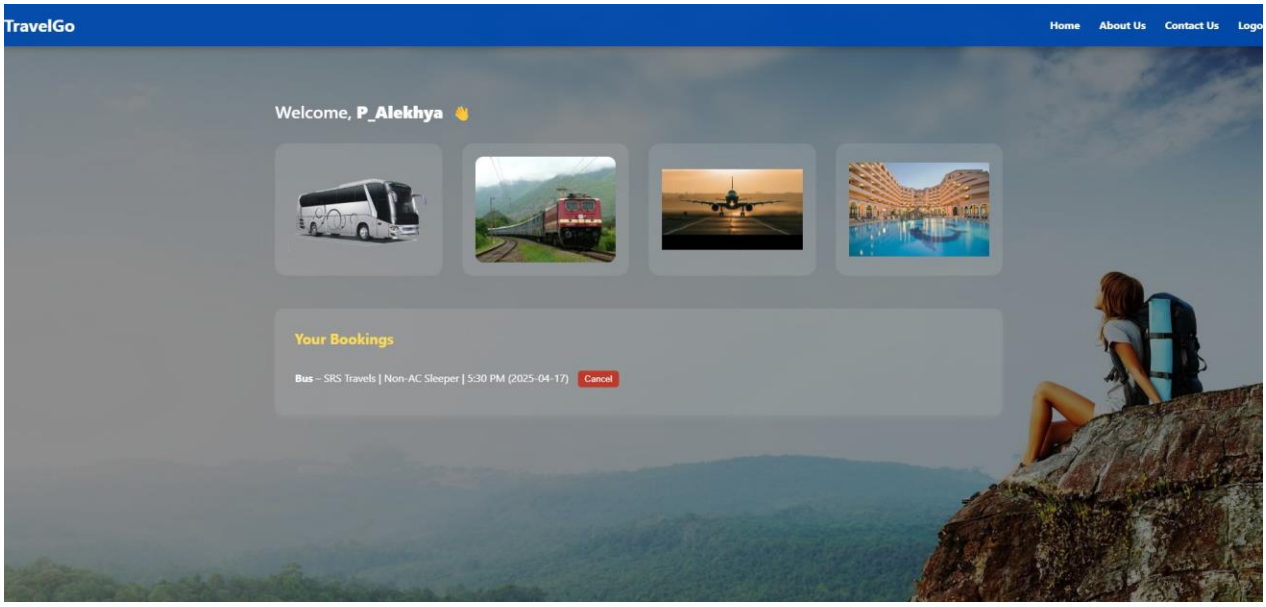

Create a TravelGo Account
Full Name
Email Address
Password
Confirm Password
Register
Already have an account? Login here
← Back to Home


Select Seats for SRS Travels
Non-AC Sleeper • 5:30 PM • ₹610/person
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
Confirm Seats & Pay
* You can select only the number of seats you entered previously.

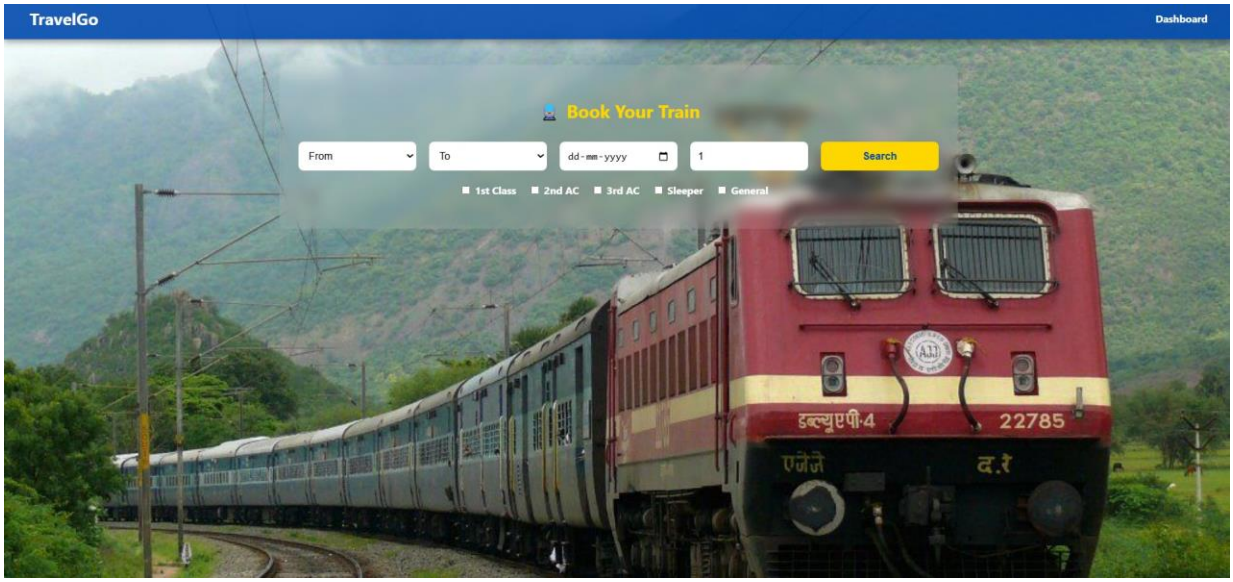


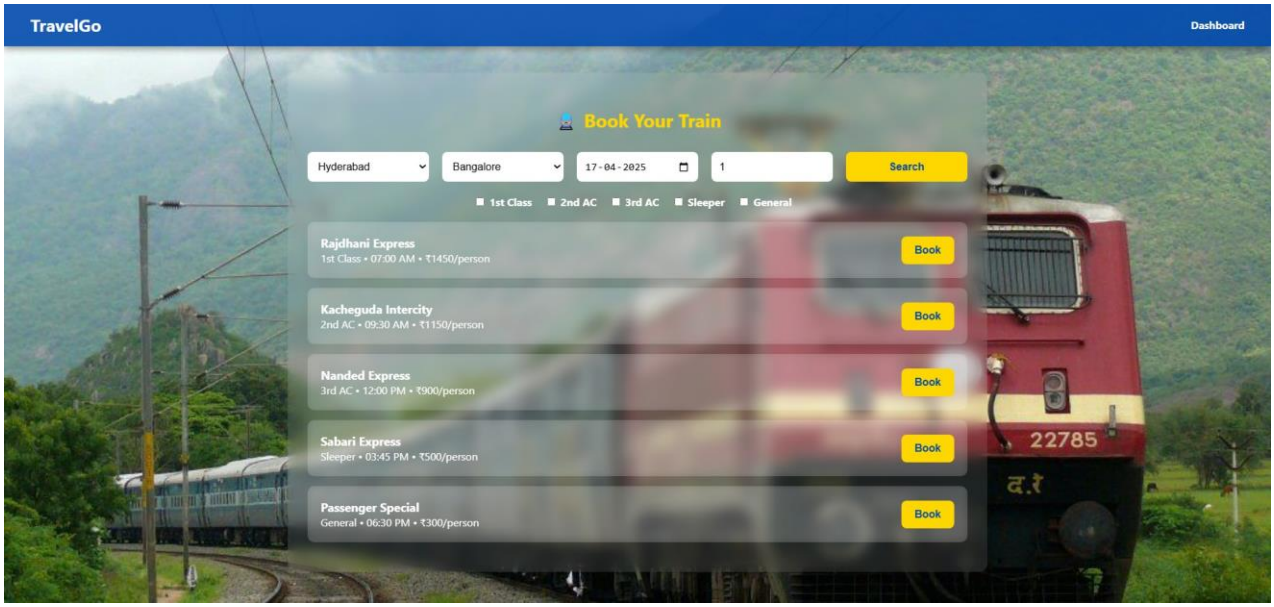
Payment Page:



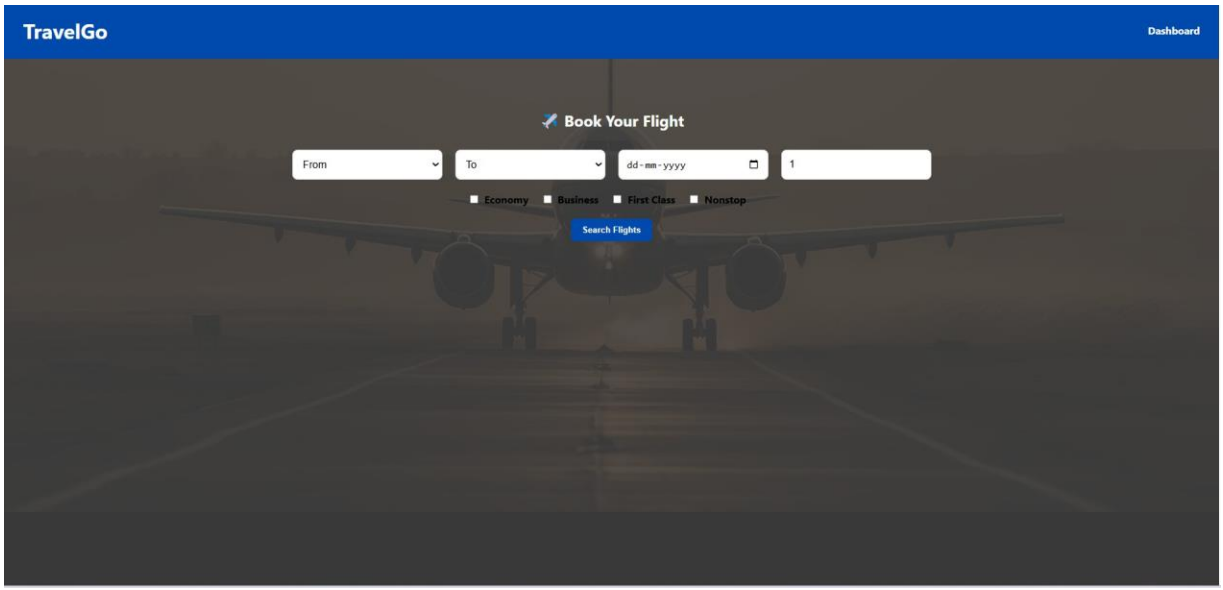

TravelGo
Home
About Us
Contact Us
Logo
Welcome, P_Alekhya
Your Bookings
Bus – SRS Travels | Non-AC Sleeper | 5:30 PM (2025-04-17)
Cancel

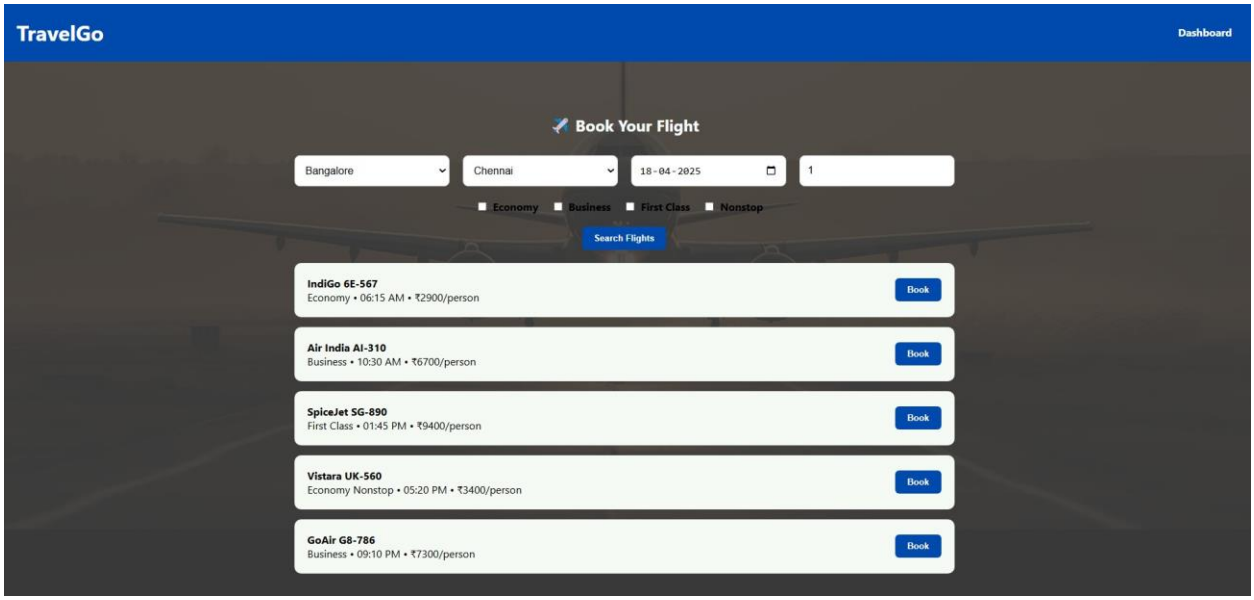
Trains page:


TravelGo
Dashboard
Book Your Train
From
To
dd-mm-yyyy
1
Search
1st Class
2nd AC
3rd AC
Sleeper
General
22785

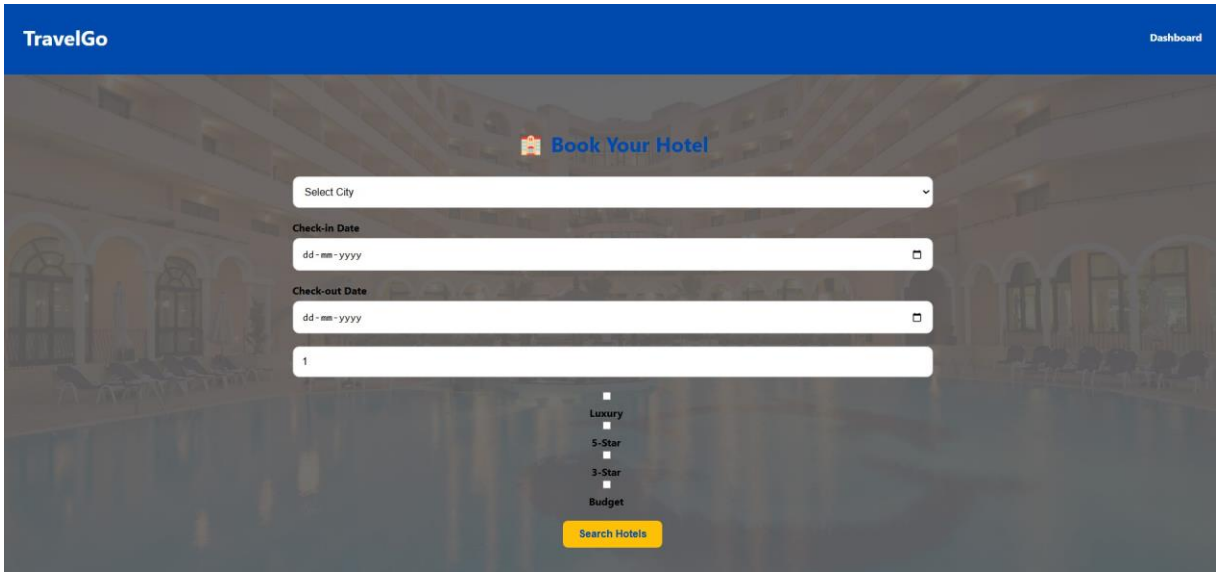


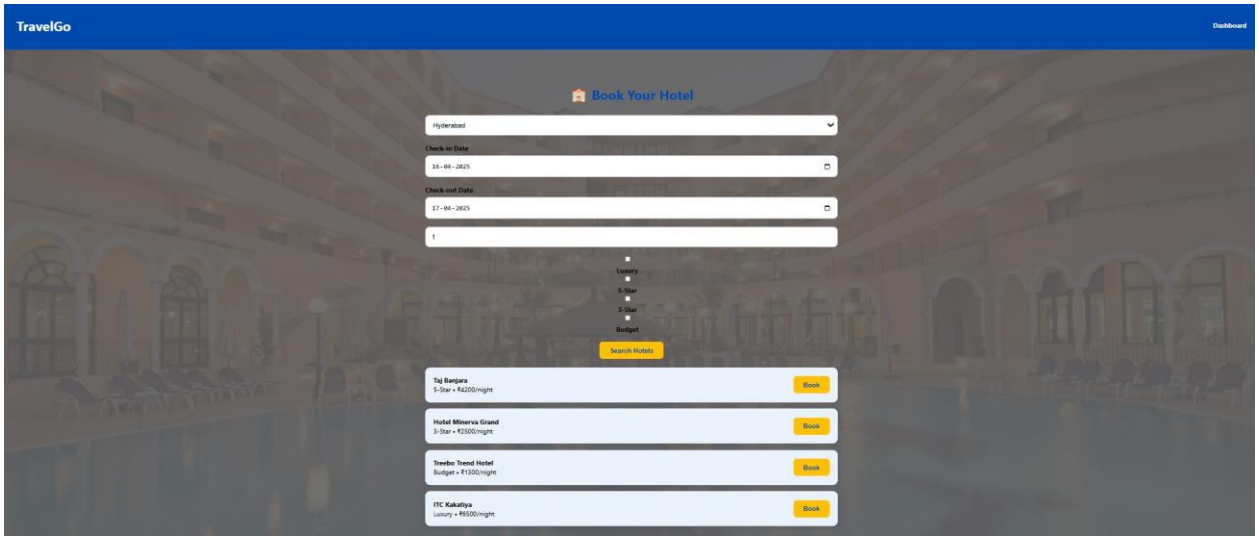
Flights Page:



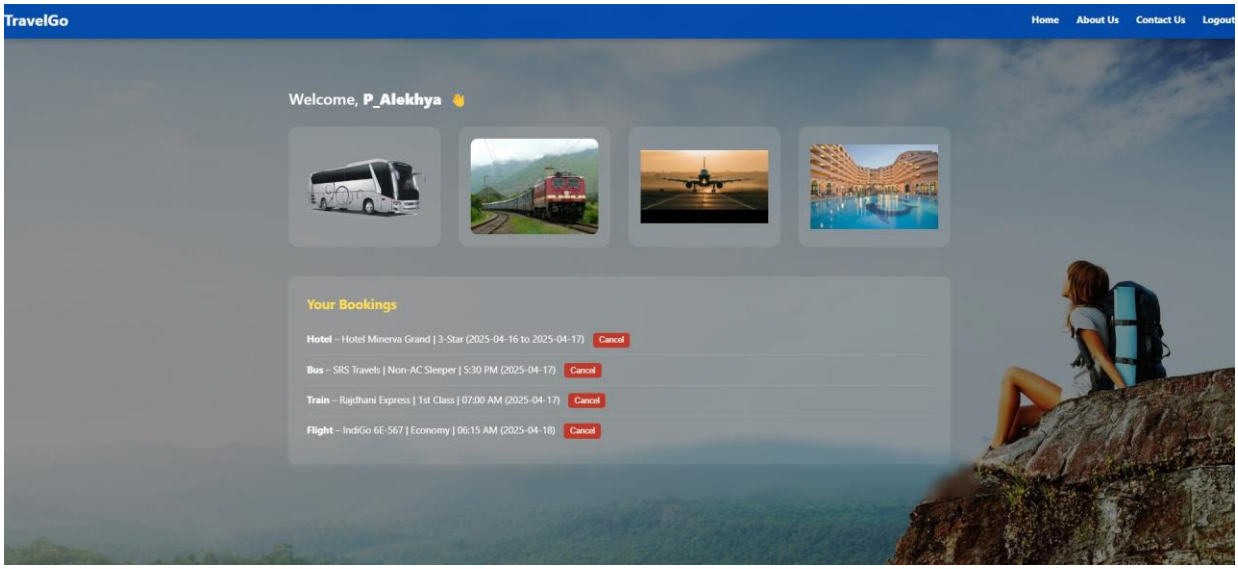

TravelGo
Dashboard
Book Your Flight
Bangalore
Chennai
18-04-2025
1
Economy
Business
First Class
Nonstop
Search Flights
IndiGo 6E-567
Economy • 06:15 AM • ₹2900/person
Book
Air India AI-310
Business • 10:30 AM • ₹6700/person
Book
SpiceJet SG-890
First Class • 01:45 PM • ₹9400/person
Book
Vistara UK-560
Economy Nonstop • 05:20 PM • ₹3400/person
Book
GoAir G8-786
Business • 09:10 PM • ₹7300/person
Book

Hotels Page:


TravelGo
Dashboard
Book Your Hotel
Select City
Check-in Date
dd-mm-yyyy
Check-out Date
dd-mm-yyyy
1
Luxury
5-Star
3-Star
Budget
Search Hotels



Dashboard with bookings Page:



Dynamodb Database updations :

1. **travel-users table :**

Table: travel-Users - Items returned (3)

Scan started on April 18, 2025, 11:21:41

Actions ▼ Create item

1

email *(String)*	logins	name	password
kilarukusuma12@gm...	1	KILARU KU...	123
sunder@thesmartbrid...	1	Sunder	12345
alekhya080228@gma...	3	P_Alekhya	1234

Description: Store and manage user credentials, login counts, and names in the `travel-Users` table on DynamoDB for authentication and tracking purposes.

2. **Mail to the User:**

AWS Notifications <no-reply@sns.amazonaws.com> Tue, Apr 15, 9:11PM (3 days ago)

to me

✅ Booking Confirmed!
SRS Travels | Non-AC Sleeper | 5:30 PM on 2025-04-17 for ₹1220

--
If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:
https://sns.ap-south-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:ap-south-1:557690616836:BookingConfirmation:da5ad931-7e8f-4ac8-b771-ec2e0cfbaf23&Endpoint=instantlibrary2@gmail.com

Please do not reply directly to this email. If you have any questions or comments regarding this email, please contact us at https://aws.amazon.com/support

Conclusion:

The **TravelGo Website** has been successfully developed and deployed using a scalable and cloud-native architecture. Leveraging AWS services such as **EC2** for hosting, **DynamoDB** for real-time data management, and **SNS** for instant booking and cancellation notifications, the platform provides a seamless travel booking experience for users. TravelGo enables registered users to search and book buses, trains, flights, and hotels in a centralized, intuitive interface, eliminating the complexities of navigating multiple travel services.

The cloud infrastructure ensures high availability and smooth performance even during peak usage, while the Flask backend ensures efficient handling of user authentication, dynamic booking flows, and data transactions. Real-time notification integration via AWS SNS allows users to receive booking confirmations and cancellations immediately via email, improving communication and user engagement.

In summary, the **TravelGo Website** offers a modern, reliable, and user-friendly solution for managing travel and accommodation needs. It highlights the potential of cloud-based platforms in building unified travel systems, simplifying operations, and enhancing the overall user experience.